Politecnico di Milano

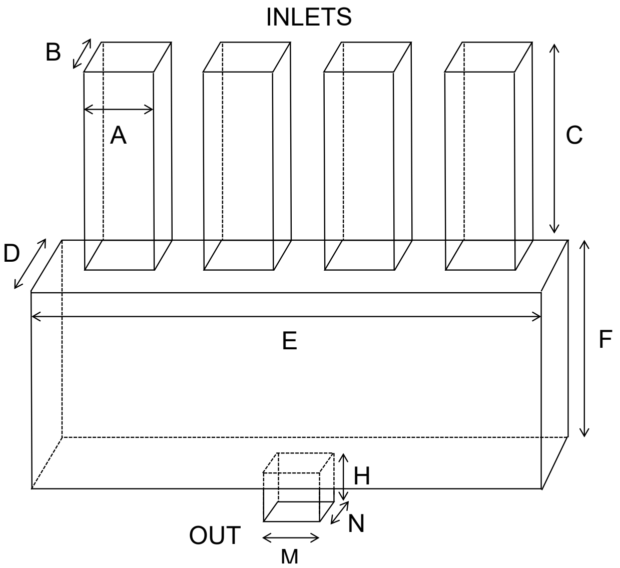
Biochip A.Y. 2023/2024 - M. Carminati

Exam of 18.07.2024

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1

Consider the following wearable microfluidic device for analysis of sweat. It consists of 4 identical inlets and one large collection chamber with sensing electrodes.



Data:

A = 100 μm
B = 20 μm
C = 2 mm
D = 500 μm
E = 5 mm
F = 2 mm
H = 100 μm
M = 200 μm
N = 200 μm

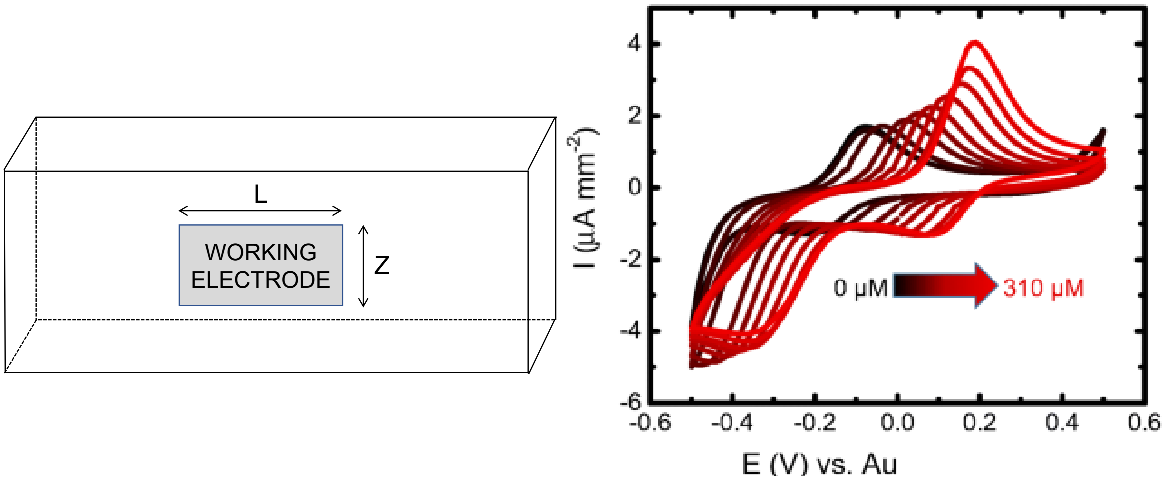
1 atm = 101 kPa

$\eta_{water} = 10^{-3}$ Pa·s
$\eta_{Air} = 1.8 \cdot 10^{-5}$ Pa·s
$\rho_{water} = 1000$ kg/m^{3-}
$\rho_{Air} = 1$ kg/m^3

a) Draw the **equivalent model** of the fluidic circuit and compute its parameters assuming it fully filled with water.
b) Assuming that a **pump** is connected to the outlet operated in aspiration with a flow date of 2 μL/min, find the water velocity in all sections (assume open inlets) and the pressure at the outlet. How long would it take to fill the collection chamber?
c) Compare a diaphragm vs. a syringe pump.
d) In order to achieve capillary flow, would it be better to have a large or a small contact angle of the inner surfaces of the channels?
e) Considering a sphere of water of 10 μm diameter, free falling by the force of gravity <u>in air</u>: what **terminal velocity** will it reach?
f) Describe the **fabrication steps** to realize the device in PDMS.

Exercise 2

Now focus on the electrochemical sensing section performed by a working electrode placed on the vertical side of the collection chamber, considering in particular the measurement of 2 parameters of the sweat sample: (a) ascorbic acid concentration by means of **cyclic voltammetry** (see figure where the max. concentration of 310 μM is shown) and (b) salinity by means of **impedance**.



Data:

V_{DD} = +5 V
V_{SS} = -5 V

L = 600 μm
Z = 200 μm

$\varepsilon_0 = 8.86 \; 10^{-12}$F/m
$C_{DL0\text{-}Au} = 0.1$pF/μm^2

a) Assuming **gold** as metal of the working electrode and <u>neglecting other electrodes</u>, draw and comment the equivalent interface impedance in the presence of ascorbic acid, computing values when possible.
b) Draw the schematic of the circuit to measure cyclic voltammetry. Size all the key values and set a bandwidth compatible with a **scan rate** of 1000 V/s.
c) What is the impact of the value of the double layer capacitance on the voltammogram?
d) Draw a modification of the circuit of point b) to measure also the sweat **conductivity**. Assuming a baseline value for the solution resistance of 20 kΩ, increasing of max. 20% among samples, draw the Bode plot of the interface impedance and select the optimal sensing frequency to measure sweat conductivity variations.
e) Study the input-referred **noise** of the circuit proposed in (d) assuming for the opamp an equivalent of 2 nV/√Hz.
f) Propose a way to measure the level of sweat filling of the collection chamber.

Multichance students with 30% reduction of the exercises please skip points 1d), 1f), 2c), 2f)